Airborne LiDAR Semantic Segmentation with Label-Free Local Geometric Context, Robust Normalization, and External Geographic Validation

Technical state and commercial MVP report

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Protocol correction (2026-07-13). All DINO/DINOv2 results and all reported mIoU values in this document are historical pre-fix figures. The DINO input channel contract was incorrect and the former IoU implementation could omit predictions of the ignore class. These figures are invalid for current comparison until caches are rebuilt and inference is repeated with `pnoa-spectral-v2` and `segmentation-metrics-v2-pred-ignore-is-fn`.

Abstract

This report describes the technical state of an airborne LiDAR semantic-segmentation and forest-intelligence minimum viable product before the 2026-07-13 protocol correction. The deployable core is a supervised point-wise model enhanced with label-free multi-scale geometric context, Taubin–Weingarten-inspired descriptors, robust coordinate and spectral normalization, focal loss, and class-balanced training. Historical artifacts retained two models for different objectives and reported overall accuracy, macro-F1, and mean-IoU figures throughout this document. Those figures are preserved only as pre-fix provenance: the mIoU values and every comparison involving DINO must be regenerated under the corrected feature and metric schemas before they can support a current ranking or deployment claim. Water and building support were also sparse in the fresh holdout, so absolute external claims for those classes require a more representative dataset. Geo-JEPA and TW-JEPA remain implemented research tracks rather than demonstrated sources of the strongest downstream result. The product code delivers GIS-ready semantic layers, point-cloud exports, forest proxies, anomaly candidates, and auditable evaluation reports; corrected benchmark results remain pending.

Contents

1 Executive Summary

The project converts airborne LiDAR point clouds into operational land-cover and forest-intelligence outputs. Its six semantic classes are:

1. ground;
2. low vegetation;
3. medium vegetation;
4. high vegetation;
5. building;
6. water.

The current product should be described as:

> A supervised airborne LiDAR semantic-segmentation and forest-intelligence platform enhanced with label-free local geometry and robust normalization, with strong internal performance and demonstrated improvement over a strong baseline on a separate geographic holdout.

The project now has two relevant model references:

- **Internal macro-F1 and mIoU leader:** `geom_concat_h256_balancedval`.
- **Deployment and geographic-generalization candidate:** `geom_robustnorm_cb20k_seed42`.

The distinction is important. The first model has the highest internal macro-F1 and mIoU. The second keeps almost the same internal quality, obtains the highest internal overall accuracy, and performs materially better under geographic and sensor-domain shift.

2 Business Motivation

Airborne LiDAR provides explicit 3D measurements of terrain, vegetation, buildings, and water. Regional surveys, however, may contain hundreds of millions or billions of points. Manual interpretation is slow, expensive, difficult to reproduce, and poorly suited to repeated monitoring.

The commercial value is not merely point classification. The system turns raw point clouds into:

- classified point-cloud layers;
- GIS-ready terrain, vegetation, building, and water maps;
- forest structure and canopy proxies;
- anomaly and disagreement candidates;
- prioritized review areas;
- reproducible metrics and technical reports.

A customer can begin with a bounded pilot on one area and one operational use case. The result is a measurable decision about scaling, rather than a large up-front platform commitment.

3 Current Product Scope

3.1 Deployable segmentation core

The deployable route combines:

- supervised point-wise segmentation;
- Taubin–Weingarten-inspired surface descriptors;
- a 56-dimensional label-free geometric-context cache;
- local statistics at 2.5 m, 5 m, and 10 m;
- focal loss;
- inverse-square-root class weighting;
- class-balanced block sampling;
- robust coordinate and spectral normalization.

The local-context cache declares `uses_labels=false`. It is derived from coordinates, raw point features, and TW descriptors, not from ground-truth labels or test predictions.

3.2 Geo-JEPA research track

Geo-JEPA is the self-supervised research direction. Its intended task hides spatial regions and predicts latent target representations from visible context. Implemented components include:

- LiDAR/PNOA data loaders and COL/CIR pairing;
- spatial masking;
- GeoPointJEPA;
- TW-JEPA auxiliary prediction;
- latent embedding prediction;
- SIGReg-style regularization;
- frozen, semi-frozen, and fine-tuned downstream evaluation.

Frozen JEPA encoders did not beat the strongest supervised routes. Geo-JEPA therefore remains a research track for future pretraining, domain adaptation, and uncertainty-aware representation learning.

3.3 Forest intelligence layer

The forest-facing layer derives operational indicators from semantic predictions and LiDAR geometry:

- vegetation ratio;
- high-vegetation ratio;
- forest-core ratio;
- canopy-cover proxy;

- canopy-height proxy;
- canopy-gap flags;
- per-cell and per-tile metrics;
- rule-based anomaly candidates.

These outputs are practical proxies. They are not certified biomass estimates, species classifications, or verified temporal-change measurements.

4 Data and Experimental Protocol

4.1 Internal data

The internal development pipeline uses airborne LiDAR data with XYZ coordinates, RGB/CIR/NIR information, laser intensity, classification labels, and TW descriptors. The original strongest local-context run used 12,000 class-balanced training blocks, 2,000 class-balanced validation blocks, and the full 34,577-block internal test split.

The internal test split was not truncated for the reported main result. Train and validation selection were balanced to provide meaningful support for minority classes, especially buildings and low/medium vegetation.

4.2 Robust training configuration

The deployment-oriented model `geom_robustnorm_cb20k_seed42` uses:

- 20,000 class-balanced training blocks;
- 4,000 class-balanced validation blocks;
- 56 external geometric-context channels;
- TW input features;
- `coordinate_normalization=xy_unit_z_robust`;
- `spectral_normalization=block_robust`;
- robust normalization of external spectral context;
- focal loss and class-balanced sampling;
- 35 requested epochs and 33 completed epochs;
- early stopping;
- best validation macro-F1 of 0.816798.

4.3 External development benchmark

The first external CAT-32 set contains 32 geographically separate `CAT-2016` COL/CIR tile pairs:

- 48,696 blocks;
- 119,005,770 points;
- all six semantic classes;

- Galicia campaign prefixes excluded;
- training-pipeline TW normalization reused;
- geometric context generated without labels.

This set was initially used as an external evaluation set. It was subsequently used to diagnose domain shift and choose the robust-normalization strategy. It is therefore reported as an **external development and robustness benchmark**, not as the final untouched holdout.

4.4 Fresh tile-disjoint external holdout

A second set was selected without tile overlap with the external development benchmark:

- 32 `CAT-2016` tiles;
- 49,853 blocks;
- 130,588,254 points;
- 51.76% ground;
- 8.80% low vegetation;
- 10.51% medium vegetation;
- 28.07% high vegetation;
- 0.80% building;
- 0.064% water.

This holdout is geographically separate and tile-disjoint from the first CAT-32 set. It is the strongest current evidence of out-of-region generalization. Its main limitation is inadequate support for water and limited support for buildings.

5 Feature Representation

5.1 Taubin–Weingarten descriptors

TW descriptors provide physically meaningful local surface information. In the supervised route, they expose shape and curvature cues beyond raw XYZ and color values. This is useful for separating flat roofs, continuous terrain, irregular vegetation, and local boundaries.

5.2 Multi-scale local geometric context

The geometric-context cache contains 56 features derived at 2.5 m, 5 m, and 10 m scales. Signals include:

- point density and local counts;
- local height range and relative height;
- coordinate summaries;
- intensity and color summaries;
- NIR and NDVI-like cues when available;

- TW summaries and local surface statistics.

The intuition is straightforward. A single high point may be ambiguous, but an irregular high neighborhood is likely a tree crown, while a flat rectangular neighborhood is more likely a roof.

5.3 Robust normalization

The first local-context model remained strong internally but degraded under external domain shift. The corrected route reduces sensitivity to region-specific photometric and height distributions by using robust within-block normalization for coordinates and spectral channels.

6 Internal Results

6.1 Best internal macro-F1 and mIoU run

Table 1 compares the original internal baseline with `geom_concat_h256_balancedval`.

Table 1: Internal test metrics for the original baseline and best internal mIoU run.

Metric	Original baseline	Local-context model	Delta
Overall accuracy	83.03%	86.49%	**+3.45 pp**
Macro-F1	74.82%	82.60%	**+7.78 pp**
Mean IoU	63.49%	72.63%	**+9.14 pp**

Table 2 shows that all six class IoUs improve.

Table 2: Per-class internal IoU improvement for the best internal mIoU run.

Class	Baseline IoU	Best IoU	Delta
Ground	72.63%	75.60%	**+2.97 pp**
Low vegetation	33.54%	44.39%	**+10.85 pp**
Medium vegetation	40.70%	53.29%	**+12.59 pp**
High vegetation	91.06%	94.74%	**+3.69 pp**
Building	45.75%	69.88%	**+24.13 pp**
Water	97.26%	97.88%	**+0.62 pp**

6.2 Three-seed stability

All three seeds were positive against the original internal baseline and improved the tracked class IoUs.

6.3 Robust deployment candidate

The robust model reaches 86.71% overall accuracy, 82.48% macro-F1, and 72.50% mIoU internally. It has the highest internal overall accuracy and nearly matches the original mIoU leader.

Table 3: Three-seed internal stability of the original local-context configuration.

Run	OA	Macro-F1	mIoU
Seed 42	86.49%	82.60%	72.63%
Seed 1337	85.92%	82.00%	71.87%
Seed 2026	84.61%	79.97%	69.52%
Mean	85.67%	81.52%	71.34%
Worst run	84.61%	79.97%	69.52%

Table 4: Internal per-class metrics for the robust deployment candidate.

Class	Precision	Recall	F1	IoU
Ground	83.72%	89.86%	86.68%	76.49%
Low vegetation	67.98%	55.36%	61.02%	43.91%
Medium vegetation	64.72%	73.96%	69.03%	52.71%
High vegetation	98.79%	95.63%	97.19%	94.53%
Building	76.01%	89.05%	82.01%	69.51%
Water	98.55%	99.29%	98.92%	97.86%

7 External Geographic Validation

7.1 External development benchmark

Table 5 reports the first CAT-32 external benchmark. The robust model improves all six class IoUs against the strong TW baseline.

Table 5: External development benchmark on the first CAT-32 set.

Model	OA	Macro-F1	mIoU	Delta mIoU
Strong TW baseline	72.60%	63.99%	49.45%	–
Robust local-context	80.83%	73.42%	60.41%	**+10.96 pp**

The corresponding gains are +8.23 percentage points in overall accuracy, +9.43 points in macro-F1, and +10.96 points in mean IoU.

7.2 Fresh disjoint external holdout

Table 6 reports the fresh holdout. The robust model improves every global metric and all six class IoUs against the strong baseline.

The improvement is operationally meaningful, but the class composition must be interpreted carefully. Water represents only 0.064% of reliable points and buildings 0.80%. The external water IoU is therefore not representative enough for a strong absolute customer claim. An additional external set with meaningful water and building support is required.

Table 6: Global metrics on the fresh tile-disjoint external holdout.

Model	OA	Macro-F1	mIoU	mIoU excl. water
Strong TW baseline	77.33%	51.38%	40.62%	48.33%
Robust local-context	83.27%	61.80%	50.98%	60.62%
Delta	**+5.94 pp**	**+10.42 pp**	**+10.35 pp**	**+12.29 pp**

Table 7: Per-class IoU on the fresh external holdout.

Class	Strong baseline	Robust model	Delta
Ground	73.43%	79.97%	**+6.54 pp**
Low vegetation	21.23%	27.70%	**+6.47 pp**
Medium vegetation	47.28%	56.46%	**+9.18 pp**
High vegetation	81.93%	88.47%	**+6.54 pp**
Building	17.80%	50.50%	**+32.70 pp**
Water	2.09%	2.77%	**+0.68 pp**

8 Domain-Shift Analysis

The degradation of the first local-context model outside the original region was not explained solely by insufficient model capacity. The external campaign differs substantially in photometry, elevation range, density, and class priors.

Key observations include:

- mean NIR changed from 0.5050 internally to 0.2955 externally;
- NIR Jensen–Shannon divergence was 0.2420;
- mean block z-range changed from 16.99 to 61.58;
- z-range p95 changed from 42.54 to 423.26;
- class proportions changed substantially across regions.

The most effective correction was the combination of robust normalization, label-free local geometry, and genuine class-balanced selection over 20,000 training blocks. Strong spectral augmentation and metric-height variants did not outperform this configuration.

9 JEPA and Image-Foundation-Model Findings

9.1 Geo-JEPA and TW-JEPA

The repository contains working self-supervised components, but frozen downstream JEPA models remained below the strong supervised baseline. The current product should therefore not be represented as a self-supervised JEPA winner.

The most promising future JEPA direction is to place self-supervised prediction over a stronger local/contextual 3D backbone and evaluate frozen, semi-frozen, and target-domain adaptation settings under a new untouched holdout.

9.2 DINOv2

DINOv2 dense features were fused with LiDAR/TW inputs. Direct concatenation and gated residual fusion were both negative against the strong LiDAR baseline. DINOv2 is therefore not a current product improvement.

9.3 DINOv3 and V-JEPA-style extensions

Potential future experiments include:

- aerial or satellite DINOv3 late fusion;
- teacher distillation into the 3D model;
- auxiliary raster decoding;
- dense predictive losses over point and raster tokens;
- multi-date predictive modeling when aligned temporal data exists.

10 Forest Analytics and GIS Deliverables

10.1 Forest proxies

The forest layer can compute:

- point counts and height statistics;
- vegetation, high-vegetation, and medium-high ratios;
- canopy-cover proxies;
- canopy-height proxies;
- canopy-gap flags;
- dominant class and per-cell error fields;
- rule-based anomaly candidates.

When a local ground proxy is available, canopy height is estimated from vegetation z95 relative to local ground. Otherwise, a conservative z95-minus-zmin fallback is used. This must remain documented as a proxy.

10.2 Customer deliverables

A pilot or deployment can deliver:

- classified and colored LAZ files;
- terrain, vegetation, building, and water layers;
- CSV, raster, or GeoPackage-style grids;
- forest proxy maps;
- anomaly and model-disagreement maps;

- overall and class-level metrics;
- confusion matrices;
- technical and executive reports;
- a prioritized review queue.

Outputs can be integrated with QGIS, ArcGIS, PostGIS, GeoServer, internal viewers, or customer APIs.

11 Commercial Deployment Path

A low-risk commercial path is:

1. **Discover:** select one territory and one operational objective.
2. **Prepare:** audit LiDAR, reference maps, identifiers, and output requirements.
3. **Pilot:** process a representative sample and deliver maps and metrics.
4. **Validate:** review representative zones with GIS or field experts.
5. **Scale:** run tiled inference and merge outputs for a larger territory.
6. **Operate:** monitor data quality, model drift, and future survey updates.

The product's commercial promise is operational acceleration: less manual mapping, faster review, consistent geographic layers, and measurable evidence before full integration.

12 Limitations

The current limitations are:

- the deployable winner is supervised rather than a self-supervised JEPA model;
- low vegetation remains the weakest class;
- the fresh external holdout has insufficient water support and limited building support;
- calibrated uncertainty is not yet a completed production component;
- complete production-scale tiled inference and merging still require hardening;
- forest outputs are proxies rather than certified inventory variables;
- true temporal change detection requires aligned multi-date data;
- a modern strong local 3D backbone has not yet been reproduced under the same complete protocol.

These limitations do not invalidate the current MVP. They define the correct scope for a paid technical pilot and the evidence required for broader claims.

13 Recommended Next Steps

1. obtain another external holdout with meaningful water and building support;

2. benchmark an efficient local 3D backbone such as a RandLA-, Point-Transformer-, KPConv-, or Superpoint-Transformer-style model;
3. add calibrated confidence and uncertainty outputs;
4. productionize tiled inference, merging, and GIS packaging;
5. validate forest proxies against authoritative or field measurements;
6. continue Geo-JEPA over the robust local-context backbone;
7. document any unlabeled target-domain adaptation as transductive domain adaptation and retain another untouched holdout;
8. add multi-date analysis only when aligned temporal surveys are available.

14 Conclusion

The project has moved beyond an internal-only LiDAR experiment. It now has:

- a strong internal segmentation result;
- a robustness-oriented deployment candidate;
- a diagnosed and corrected domain-shift problem;
- clear improvement over a strong baseline on a fresh geographic holdout;
- GIS-ready outputs and forest analytics;
- a transparent distinction between deployable supervised performance and the ongoing Geo-JEPA research track.

The strongest concise statement is:

> This MVP turns airborne LiDAR into GIS-ready land-cover maps, forest indicators, and prioritized review areas. Its robust supervised model combines label-free local geometry with domain-aware normalization and demonstrates clear improvement over a strong baseline on geographically separate, tile-disjoint data.